Episode 9: The Pitch Deck

Programming with Python · Tutorial 09



The pitch meeting is Friday, and the investor gave one instruction: **"Bring me charts I can't argue with."** Last week you turned the data room into numbers she could trust; this week those numbers have to be *seen*. Same file, `orders_messy.csv`, now with pictures: you pick the chart that fits each question, save it as a picture file, and end the lab with a deck of six. Tobi has already "built the deck": two slides and one argument for a new rider. Your job is to **supervise**: catch the chart she can argue with before she does.

Run the lab on your laptop

The lab is a Python script again. It runs in the `python-labs` folder, next to last week's files.

1. Download the script and move it into your `python-labs` folder. `orders_messy.csv` is still there from last week; if not, download it again:

   Lab script Orders data

2. In Zed, open the `python-labs` folder, then the terminal (`Ctrl+`).
3. Run `uv run python lab_09_pitch.py`.

> **No Zed on your laptop, or only an iPad?**
> Use a **GitHub codespace**: a Linux computer that runs in a browser tab.
>
> 1. Sign in at github.com/codespaces and open last week's codespace again. If you have none, start the **Blank** template and follow the uv guide (the macOS or Linux lines) in its terminal, down to `uv add pandas matplotlib`.
> 2. Go into the folder with `cd python-labs` and download the script. The second line is only needed in a new codespace:
>
>    ```
>    curl -O https://python.tobiasvlcek.com/labs/lab_09_pitch.py
>    curl -O https://python.tobiasvlcek.com/labs/orders_messy.csv
>    ```
>
> 3. Run `uv run python lab_09_pitch.py`.
>
> To look at a chart, click its PNG file in the file list on the left. A codespace has no Zed, so keep the chatbot open in a second tab for AI help.

The first run stops with `AssertionError: Task 1: not attempted yet.`, as it did last week: the script checks every task and stops at the first one you have not solved yet. The very first run can take a little longer, because matplotlib sets up its fonts once.

If the run ends in `ModuleNotFoundError` instead, the terminal is not inside `python-labs`, or the packages are missing. In that folder, run `uv add pandas matplotlib` once.

A script has no cell that could show a chart, so every chart goes into a picture file in `python-labs`. Task 2 in the script shows the three lines that end every chart. To look at a picture, click its file in Zed's project panel, the file list on the left.

> ! **Look at your charts**
> The checks only test the form of a chart: that it has a title, that its bars add up to your own numbers. They cannot see whether the chart tells the truth. And **two things in Tobi's deck are wrong without any task saying so**: one is in his numbers, one is in what he claims. No check catches either of them.

When every check passes, the script puts your six charts side by side into `pitch_deck.png` and prints a block that starts with `SHOW THIS IN CLASS`. Show both to your lecturer. As in lab 08, that is feedback, not a condition for leaving, and a lab you finish at home does not have to be shown later.

Checkpoint 5 next week is still handed in from the browser, and the script labs no longer practise that. So take two minutes before you go: open today's last exercise, ex_09_e, press **Cmd/Ctrl+S**, then menu → Download → **Download Python code**, and find the file in your Downloads folder.